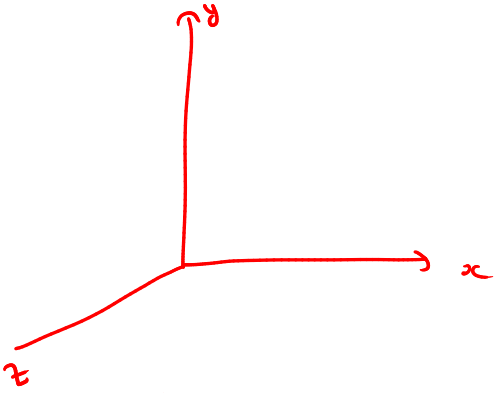
Vector ✓

Vector = $\begin{bmatrix} 3 \\ 5 \end{bmatrix}$

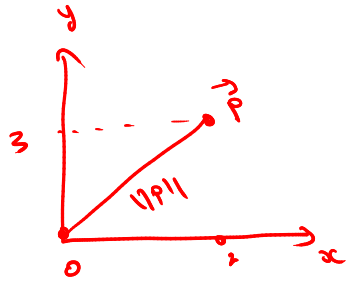
Why vector — mathematical

"How is the weather today" ⇒ Embedding ⇒ vector $\begin{bmatrix} \cdot \\ \cdot \\ \cdot \end{bmatrix}$

vector → magnitude & Direction

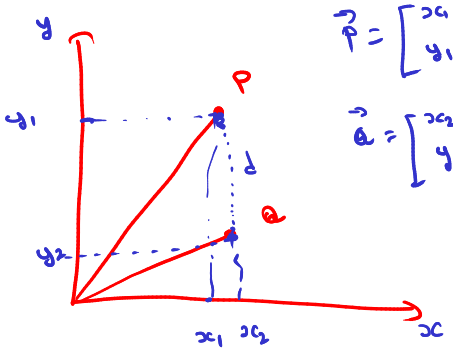


$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$



||P|| - length of vecto

$\vec{P} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$



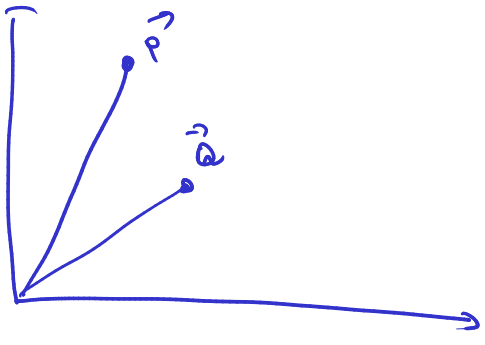
$\vec{P} = \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$

$\vec{Q} = \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$

$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$ — 2D

$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$

$d = \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}$



$\vec{P} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$ $\vec{Q} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$\vec{P} * \vec{Q} = \begin{bmatrix} 4 \\ 3 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$

$\vec{P}^T * \vec{Q} = \begin{bmatrix} 4 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = (4*3) + (3*4)$

$=$

Dot product ✓

$\vec{p} \cdot \vec{q} = p^T * q =$

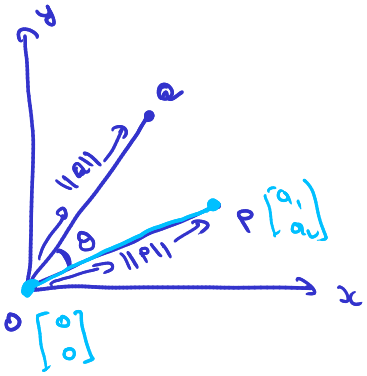
$$p = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad q = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$$p^T * q = \begin{bmatrix} a_1 & a_2 & a_3 & \cdots & a_n \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$= a_1 \cdot b_1 + a_2 b_2 + a_3 \cdot b_3 \cdots a_n b_n$

$$\boxed{\vec{p} \cdot \vec{q} = p^T * q = \sum_{i=1}^{n} a_i b_i}$$ ← Dot product

Geometry view



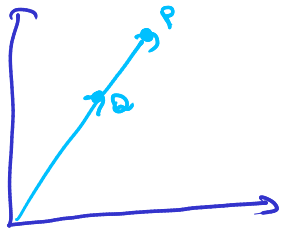
$\vec{p} \cdot \vec{q} = \|p\| \cdot \|q\| \cdot \cos\theta = p^T q = \sum_{i=1}^{n} a_i b_i$

$\theta = 90^\circ \quad \vec{p} \perp \vec{q}$



$\cos 90 = 0$

$$\boxed{\vec{p} \cdot \vec{q} = 0}$$



$\theta = 0$

$\cos\theta = 1$

$\vec{p} \cdot \vec{q} = \|p\| \cdot \|q\|$

$\|p\| =$

$$p = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad q = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$

$\|p\| = \sqrt{\sum_{i=1}^{n} a_i^2}$

$\|p\| = \sqrt{a_1^2 + a_2^2 + a_3^2 + \cdots a_n^2}$

$= \sqrt{\sum_{i=1}^{n} a_i^2}$

$\|q\| = \sqrt{\sum_{i=1}^{n} b_i^2}$

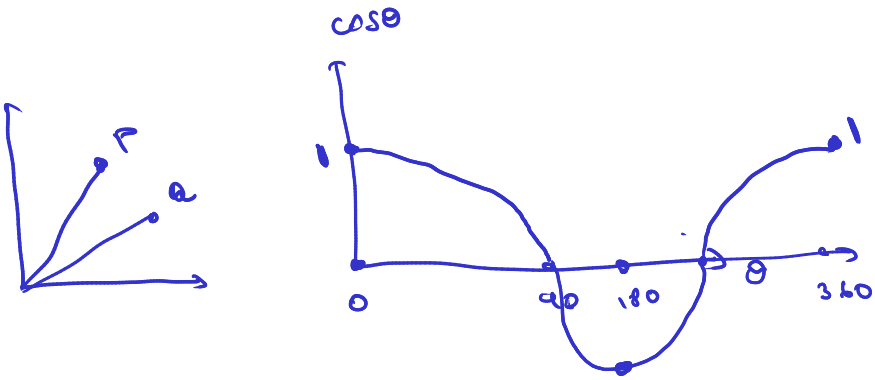
$\vec{p} \cdot \vec{q} = \|p\| \cdot \|q\| \cdot \cos\theta$

$$\cos\theta = \frac{\vec{p} \cdot \vec{q}}{\|p\| \cdot \|q\|}$$

← cosine similarity — 0 to 1

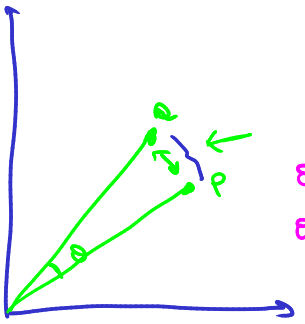
↓

Angular Distance



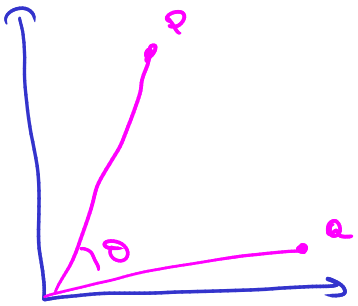
θ ↓ cosθ ↑

θ ↑ cosθ ↓



θ ↓ cosθ ↑

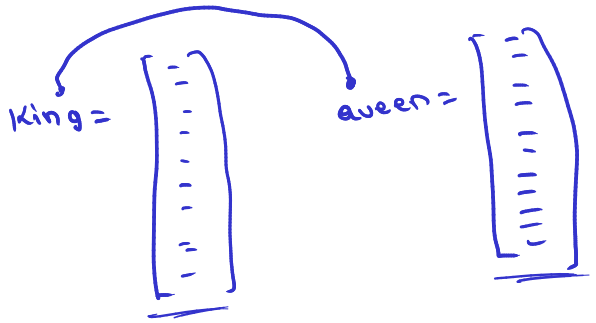
θ = 10 cos = 0.98

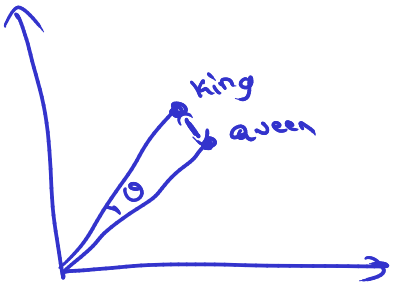


θ = 60 cosθ = 0.5

$$\cos\theta = \frac{\vec{p} \cdot \vec{q}}{\|p\| \cdot \|q\|} = \frac{\sum_{i=1}^{n} a_i b_i}{\sqrt{\sum_{i=1}^{n} a_i} \cdot \sqrt{\sum_{i=1}^{n} b_i}}$$

$$\vec{p} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \quad \vec{q} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}$$





cosθ = high

Distance = low

RAG

1. Cosine similarity
2. Euclidean Distance

→ MMR